SRM | Reg No. | R | A | 2 | 4 | 1 | 2 | 7 | 0 | 3 | 0 | 1 | 0 | 0 | 0 | 7 |

SET E

SRM Institute of Science and Technology

College of Engineering and Technology

SCHOOL OF COMPUTING

ASSESMENT II

Course Code and Title: 21CSC203P / Advanced programming practice **Year / Sem: II / III**

Max. Marks: 25 **Duration: 1 Hr** **Date : 03.11.2025**

COURSE ARTICULATION MATRIX (CAM)

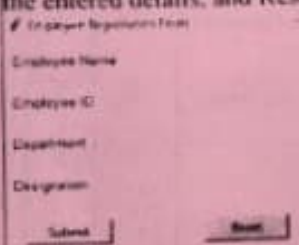
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	3	-	-	-	-	-	-	-	-	-	-
CO2	3	2	-	1	3	-	-	-	-	-	-	-	-	-	-
CO3	3	-	2	1	3	2	-	-	1	-	-	-	-	-	-
CO4	3	2	-	1	3	-	-	-	-	-	-	-	-	-	-
CO5	3	-	2	1	3	2	-	-	1	-	-	-	-	-	-

Sl. No	Part A (5* 2 = 10 Marks) (Answer ALL Questions)	Marks	CO	PO	BL	PI CODE
1.	Define a pure function. Write a Python program using a pure function and a lambda expression to double all even numbers in the list [1, 2, 3, 4, 5, 6] and display the resulting list.	2	4	1	1	4.1.1
2.	Demonstrate how a Lock() prevents data corruption in a shared list using Python's threading module.	2	4	3	4	4.3.2
3.	Write a Python statement to create a TCP socket and to bind a server socket to localhost on port 5000.	2	4	3	3	4.3.3
4.	Write a Python program to substitute x=2 in the expression x^2+3x using SymPy.	2	5	3	2	5.2.2
5.	Consider an automatic door with the following specifications: • States: CLOSED, OPEN • Inputs: person_detected, no_person. Draw the state transition table for the automatic door.	2	5	3	3	5.2.2

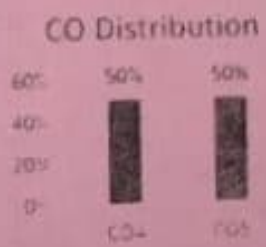
Part B (3* 5 = 15 Marks) (Answer ALL Questions)

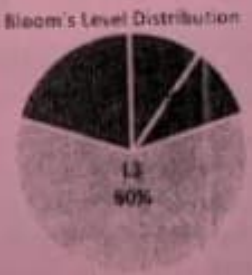
SLNo	Question	Marks	CO	PO	BL	PI Code
Q6 a.	A company has a list of employee salaries: [2000, 3500, 4000, 1500, 5000]. Write a Python program to: 1. Apply a 5% bonus to each salary using map(). 2. Keep only salaries above 3000 using filter(). 3. Calculate the total sum of the filtered salaries using reduce(). Display the original salaries, updated salaries, filtered salaries, and total.	5	4	3	3	4.3.4
	(OR)					
Q6 b.	Develop a Python TCP client-server application using the socket module where the server listens on localhost:5000, receives a client message, and sends a confirmation reply. The client should take user input, send it to the server, display the server's response, and ensure proper socket closure on both ends.	5	4	3	3	4.3.4

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Q7 a.	Differentiate between shared and distributed memory models. Illustrate your answer with diagrams and short Python programs demonstrating each concept.	5	5	3	3	4.1.1
	(OR)					
Q7 b.	Consider the following facts about students and the clubs they have joined: student('David'). student('Emma'). student('Frank'). club('Chess'). club('Robotics'). club('Music'). joined('David', 'Chess'). joined('Emma', 'Music'). joined('Frank', 'Robotics'). joined('David', 'Music'). 1.Write a Prolog or PyDatalog query to list the names of students who have joined the 'Music' club. 2.Write a Prolog or PyDatalog query to list all student–club membership pairs stored in the database. 3.Write a Prolog or PyDatalog query to find all clubs that 'David' has joined. 4. Write a Prolog or PyDatalog query to list students who are not enrolled in the 'Robotics' club.	5	4	3	3	4.1.1
Q8 a.	Develop a Python program using the SymPy library to compute both the derivative and integral of the given mathematical expression: $f(x)=3x^3-5x^2+2x-7$	5	5	3	3	5.3.2
	(OR)					
Q8 b.	Develop a Python GUI program using Tkinter to create an Employee Registration Form with fields for Name, ID, Department, and Designation. Include Submit and Reset buttons — Submit displays the entered details, and Reset clears all fields. 	5	5	1	3	5.3.2

Course Outcome (CO) and Bloom's level (BL) Coverage in Questions





Course Coordinator

Audit Professor

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